Punjab School Enrollment Analysis

Comprehensive Project Documentation & Technical Architecture (1989–2022)

1. Executive Summary

This project analyzes 34 years of district-wise, gender-disaggregated school enrollment records (Classes 1–5 and Classes 6–8) in Punjab, India. The workflow integrates clean automated data engineering, unsupervised Machine Learning for reporting anomaly detection and district clustering, anomaly-aware ARIMA econometric time-series forecasting, and an interactive Streamlit web explorer.

2. Libraries and Technologies Used in the Project

Each library was deliberately chosen to serve a specific role within the analysis pipeline or user dashboard:

Library	Category	Specific Role and Purpose in This Project
pandas	Data Processing	Ingests wide CSV tables; unpivots to tidy long-format records (melt); handles missing records; computes derived aggregates (Total, Girls %, GPI); builds rolling window statistics.
numpy	Numerical Computing	Performs fast vectorized array computations, polyfit regression slopes for long-term district trend calculation, and Mean Absolute Percentage Error (MAPE) mathematical formulas.
scikit-learn	Machine Learning	Powers Isolation Forest (300 estimators) for unsupervised anomaly scoring; K-Means clustering (k=4) for district typology; StandardScaler for feature normalization; and Silhouette Score validation.
statsmodels	Time Series Modeling	Fits Autoregressive Integrated Moving Average (ARIMA 1,1,1) models on enrollment time series; validates naive vs anomaly-aware forecasts; generates 2023–2027 forecasts with 95% confidence intervals.
matplotlib	Static Visualization	Renders publication-ready high-resolution static charts (trend curves, anomaly bar charts, ARIMA validation, forecast confidence bands) saved directly into outputs/.
seaborn	Statistical Plotting	Creates divergent color-scaled District x Year Gender Parity Index heatmaps (RdYlGn) and styled multi-attribute cluster scatter plots with bubble scaling.
streamlit	Web Dashboard	Powers the interactive web explorer (punjab_dashboard/app.py) with dynamic filters, multi-district selectors, responsive KPI metric cards, and multi-view navigation tabs.
plotly	Interactive Charts	Embeds interactive charts with custom hover data tooltips, cross-filtering, and dynamic zooming within the Streamlit dashboard.
reportlab	PDF Generation	Compiles and styles this comprehensive documentation directly into a standalone, formatted PDF document.

3. Technical Architecture & Methodology

The project pipeline transforms heterogeneous government data into actionable analytical products:

Step 1: Data Cleaning & Normalization — Converts 4 separate wide tables into 3,128 standardized observations. Missing placeholders ('NA', '-', empty strings) and zero placeholders are converted to NaN.

Step 2: Metric Derivation — Computes Total Enrollment, Girls Share %, and Gender Parity Index (GPI = Girls / Boys). Parity is achieved at 1.0.

Step 3: Feature Engineering — Calculates YoY percentage changes, exact carry-over duplicate flags, and 5-year rolling standard deviation z-scores.

Step 4: Isolation Forest Anomaly Detection — Flags multi-attribute outliers with 300 estimators and 5% contamination, detecting known reporting spikes and discovering duplicate entries.

Step 5: District Typology Clustering — Normalizes long-term GPI trajectories, rates of change, and enrollment volatility, grouping the 22 eligible districts into 4 distinct behavior clusters using K-Means.

Step 6: Anomaly-Aware Time Series Forecasting — Demonstrates that handling the artificial 2011–2013 reporting spike reduces ARIMA validation forecast error from 65.6% down to 5.7%. Projections with 95% confidence intervals are forecasted for 2023–2027.

4. Key Findings & Forecast Projections

Year	Projected Enrollment	95% CI Lower Bound	95% CI Upper Bound
2023	1,732,471	1,402,997	2,061,945
2024	1,732,369	1,258,083	2,206,655
2025	1,732,313	1,144,341	2,320,286
2026	1,732,283	1,047,535	2,417,032
2027	1,732,267	961,965	2,502,569

District Typology Clusters (K-Means, k=4):

• **Cluster 0 (Core Industrial Plains)**: Amritsar, Fatehgarh Sahib, Gurdaspur, Hoshiarpur, Jalandhar, Kapurthala, Ludhiana, Moga, Rupnagar, SBS Nagar. High total enrollment, consistent GPI progress.

• **Cluster 1 (High Volatility / Fast Developing)**: Barnala, Fazilka, SAS Nagar, Sangrur, Tarn Taran. Dynamic demographic shifts and rapid expansion.

• **Cluster 2 (Rapid GPI Catch-Up)**: Bathinda, Faridkot, Ferozepur, Mansa, Patiala, Sri Muktsar Sahib. Historically low baseline GPI in 1989 that exhibited the steepest gains over three decades.

• **Cluster 3 (Sub-Mountainous Outlier)**: Pathankot. Established in 2011 with distinct geographic and scale characteristics.

5. Documented Data Irregularities & Handling

• **2017 = 2018 Carry-Forward:** In 98.9% of district/band combinations, 2017 numbers were repeated identically in 2018 in source reports. The Isolation Forest model detected this with 70.1% recall.

• **2011–2013 Volatility Artifact:** Simultaneous multi-district enrollment spike in 2011 & 2013 with a trough in 2012, reflecting an administrative census/reporting anomaly rather than demographic change.

• **Missing Year 2008:** Completely absent across all districts in source records. Preserved as NaN and interpolated for continuous time-series modeling.

• **Administrative Carve-Outs:** New districts (Malerkotla 2021, Pathankot 2011, etc.) are maintained as NaN prior to formation rather than misleading zero enrollment.

6. Project Execution Guide

To run the ML Pipeline: .\.venv\Scripts\python.exe punjab_enrollment_pipeline.py

To launch the Streamlit Dashboard: .\.venv\Scripts\python.exe -m streamlit run punjab_dashboard/app.py

Open http://localhost:8501 in your browser to access the interactive web explorer.

Key Validation Figure (Naive vs Anomaly-Aware ARIMA):

